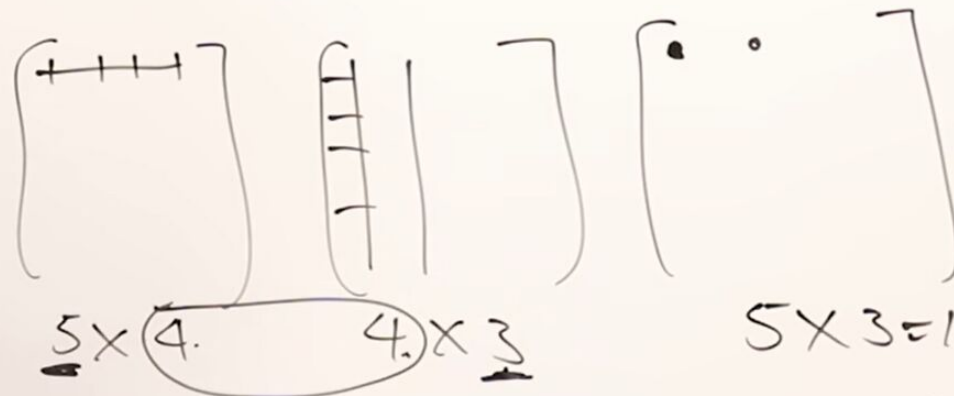


Matrix Chain Multiplication

$$\begin{array}{cccc} A_1 & \cdot & A_2 & \cdot & A_3 & \cdot & A_4 \\ 5 \times 4 & & 4 \times 6 & & 6 \times 2 & & 2 \times 7 \end{array}$$

$$A \times B = C$$



$$\begin{array}{l} B \times A \\ 4 \times 3 \leftrightarrow 5 \times 4 \end{array}$$

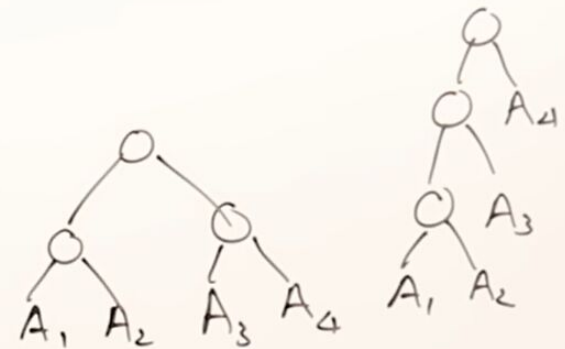
$$\begin{array}{l} 5 \times 3 = 15 \times 4 \\ 5 \times 3 \times 4 = \underline{\underline{60}} \end{array}$$

Matrix Chain Multiplication

$$\begin{array}{cccc} A_1 & \cdot & A_2 & \cdot & A_3 & \cdot & A_4 \\ 5 \times 4 & & 4 \times 6 & & 6 \times 2 & & 2 \times 7 \end{array}$$

$$((A_1 \cdot A_2) \cdot A_3) \cdot A_4$$

$$(A_1 \cdot A_2) \cdot (A_3 \cdot A_4)$$



$$T(n) = \frac{2n(n-1)}{n+1}$$

$$T(3) = \underline{\underline{5}}$$

Matrix Chain Multiplication

$$A_1 \cdot A_2 \cdot A_3 \cdot A_4$$

$$5 \times 4 \quad 4 \times 6 \quad 6 \times 2 \quad 2 \times 7$$

$$m[2,4] = 104$$

$$A_2 \cdot (A_3 \cdot A_4) \quad (A_2 \cdot A_3) \cdot A_4$$

$$4 \times 6 \quad 6 \times 2 \quad 2 \times 7 \quad 4 \times 6 \quad 6 \times 2 \quad 2 \times 7$$

$$m[2,2] + m[3,4] + 4 \times 6 \times 7 \quad m[2,3] + m[4,4] + 4 \times 2 \times 7$$

$$0 + 84 + 168 \quad 48 + 0 + 56$$

$$252 \quad 104$$

m	1	2	3	4
1	0	120	88	
2		0	48	104
3			0	84
4				0

S	1	2	3	4
1		1	1	
2			2	3
3				3
4				

Matrix Chain Multiplication

$$A_1 \cdot A_2 \cdot A_3 \cdot A_4$$

$$\begin{matrix} \textcircled{5 \times 4} & \textcircled{4 \times 6} & \textcircled{6 \times 2} & \textcircled{2 \times 7} \\ d_0 & d_1 & d_2 & d_3 & d_4 \end{matrix}$$

$$m[1,4] = \min \left\{ \begin{array}{l} m[1,1] + m[2,4] + 5 \times 4 \times 7 \\ 0 \quad 104 + 140 \end{array} , \begin{array}{l} m[1,2] + m[3,4] + 5 \times 6 \times 7 \\ 120 + 84 + 210 \end{array} , \right.$$

$$\left. \begin{array}{l} m[1,3] + m[4,4] + 5 \times 2 \times 7 \\ 88 + 0 + 70 \end{array} \right\}$$

$$m[i,j] = \min \left\{ m[i,k] + m[k+1,j] + d_{i-1} * d_k * d_j \right\}$$

m	1	2	3	4
1	0	120	88	158
2		0	48	104
3			0	84
4				0

S	1	2	3	4
1		1	1	3
2			2	3
3				3
4				

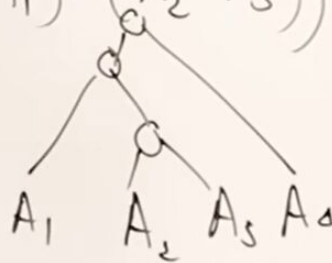
Matrix Chain Multiplication

$$A_1 \cdot A_2 \cdot A_3 \cdot A_4$$

$\underbrace{5 \times 4}_{d_0} \quad \underbrace{4 \times 6}_{d_1} \quad \underbrace{6 \times 2}_{d_2} \quad \underbrace{2 \times 7}_{d_3}$

$$(A_1 \cdot A_2 \cdot A_3) \cdot (A_4)$$

$$((A_1) \cdot (A_2 \cdot A_3)) \cdot (A_4)$$



m	1	2	3	4
1	0	120	88	158
2		0	48	104
3			0	84
4				0

S	1	2	3	4
1		1	1	3
2			2	3
3				3
4				

Matrix Chain Multiplication

$$A_1 \cdot A_2 \cdot A_3 \cdot A_4$$

$$\begin{matrix} \textcircled{5 \times 4} & \textcircled{4 \times 6} & \textcircled{6 \times 2} & \textcircled{2 \times 7} \\ d_0 & d_1 & d_2 & d_3 & d_4 \end{matrix}$$

$$\frac{n(n-1)}{2} = n^2 \times n = n^3$$

$$O(n^3)$$

m	1	2	3	4
1	0	120	88	158
2		0	48	104
3			0	84
4				0

S	1	2	3	4
1		1	1	3
2			2	3
3				3
4				

this is Theta of n cube so the time complexity for Matrix chain



$$A_1 \cdot A_2 \cdot A_3 \cdot A_4$$

$$\begin{array}{ccccc} \textcircled{5 \times 4} & \textcircled{4 \times 6} & \textcircled{6 \times 2} & \textcircled{2 \times 7} & \\ d_0 & d_1 & d_2 & d_3 & d_4 \end{array}$$

A diamond-shaped grid with numbers and circles. The grid is oriented diagonally. The numbers are: 158, 88, 104, 120, 48, 84. The circles are: 0, 0, 0, 0. The numbers are arranged in a diamond shape, with 158 at the top, 88 and 104 in the middle, 120 and 48 below that, and 84 at the bottom. The circles are arranged in a diamond shape, with 0 at the top, 0 and 0 in the middle, and 0 and 0 at the bottom.

j

m	1	2	3	4
1	○	120	88	158
2		○	48	104
3			○	84
4				○

S	1	2	3	4
1		1	1	3
2			2	3
3				3
4				

instead of drawing it like this and half unfilled or empty